

PROPOSAL FOR THE TREEQUEST CHALLENGE

Spotree

Student-led tree mapping for Munich's geospatial data infrastructure

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Theme	Students on an Urban Data Mission
Submitted by	Spotree Team
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Source Code	github.com/jayopraveen999/spotree

An educational, standards-ready mobile application for crowdsourced urban tree data collection.

Contents

1. Team

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EXPERTISE

Applied research in Earth observation, MLOps and real-time inference pipelines for satellite-based wildfire detection. **Tree species classification using satellite imagery** (university project at TUM under Prof. Xiaoxiang Zhu). Published at **ESA Living Planet Symposium 2025** and **ECAI 2020**.

Focus areas: Deep Learning, PostGIS, Geospatial Data Engineering, Python, PyTorch, Grafana, Docker, Full-Stack App Development.

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INITIATIVE · ARCHITECTURE · IMPLEMENTATION

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EXPERTISE

Publications: First author at **ICLR 2026** (HierLoc, image-based geolocation in hyperbolic space) and **SIGSPATIAL '24** (road-network matching with embeddings). Co-author at **EMNLP 2025** (Turnback, geospatial cognition of current LLMs).

Focus areas: Agentic AI, Multimodal LLMs, Proactive Agents, Deep Learning for Spatial Data, GPU Cluster Administration, Rust PostgreSQL Extensions.

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We are a two-person Munich team with complementary strengths. On one side, ML research and full-stack systems engineering. On the other, applied Earth observation and MLOps experience with real-time satellite pipelines. Both of us have been working and researching in Munich for years. We understand the local context, the TU Munich research ecosystem, and the requirements for GDPR-compliant geospatial infrastructure. The Spotree team brings over a decade of combined expertise in geospatial science, earth observation, machine learning, and spatial data infrastructure. Once selected for the co-creation phase, we are eager to refine the concept together with students and teachers, iterate on the prototype based on real-world feedback, and release a production-ready app to the students of Munich.

2. Executive Summary

Spotree (Spotify + Tree) is a mobile application that transforms urban tree mapping into an engaging, social experience for Munich's students. Instead of treating data collection as a chore, Spotree lets students "spot" trees, capture rich geospatial data, and **drop their favorite Spotify song at each tree location** — creating a musical map of Munich's urban forest.

The name says it all: **Spo(tify) + tree**. Every tree becomes a point on the map with a soundtrack. When students explore trees mapped by their peers, they can tap to play the linked song. This social, music-driven mechanic gives students a personal reason to keep mapping — they want to see their song on the map, share it with friends, and discover what others have linked. It turns a civic data project into something students genuinely want to participate in.

Behind the fun interface, Spotree builds a **production-grade geospatial database** using PostGIS with OGC-compliant spatial data types, ready for direct integration into Munich's existing geodata infrastructure.

3. Challenge Response

The challenge asks: *How can we involve students in Munich in the creation and updating of the city’s geospatial data in an active and educational way, while strengthening their connection to their neighbourhood?*

Spotree addresses every dimension of this question:

Challenge Requirement	Spotree Solution
Crowdsourced data collection	Students capture tree photos, GPS-tagged locations, species identification, health assessment, height estimation, and trunk diameter — all through a guided mobile form
Educational value	The Spotree Guide teaches 12 native Munich tree species with photos, growth characteristics, habitat info, and links to botanical references (Easyscape). Students learn to identify species before they map
Active participation	The Spotify integration creates intrinsic motivation. Students map trees to drop songs. School-level analytics enable teachers to integrate mapping into lessons
AI integration	Current prototype uses AI (LLaVA vision model) to validate uploaded photos contain trees. Co-creation phase adds AI-powered species classification
Neighbourhood connection	Students explore trees in their area, discover songs left by peers, and build a personal profile. The map becomes a living, musical portrait of their neighbourhood
Rewarding / ranking	Profile tracks trees spotted, species found, and songs shared. Co-creation phase adds leaderboards and area ownership mechanics
OGC-compliant data	All spatial data stored as PostGIS geography(Point, 4326) — standard WGS84. Export to GeoJSON, Shapefile, GeoParquet supported natively
Verification & updates	AI rejects non-tree photos. Confidence sliders on every field signal data quality. Multiple students can map the same tree, enabling cross-verification

4. App Description

Spotree is built as a native iOS app (React Native + Expo) with 5 main tabs:

4.1 Map

The central view showing all spotted trees across Munich on an interactive map. Each tree marker indicates its location; markers with a music badge have a linked Spotify song. Tapping a marker opens a detail card showing species, health, height, who mapped it, and a play button for the linked song.

[Screenshot: Map View]

4.2 Capture (Spot a Tree)

The guided data collection form. Students can:

- **Take a photo** (camera) or **pick from gallery** (with EXIF GPS extraction)
- AI validates the photo contains a tree before proceeding
- Select **tree species** from 58 Munich natives (or “Not Sure” if uncertain)
- Rate **health status** (Healthy / Fair / Poor / Dead)
- Estimate **tree height** and **trunk diameter**
- Set **confidence levels** (0–100%) for each observation via sliders
- Add **free-text notes** about the tree or its surroundings
- **Link a Spotify song** — paste a URL and name the track

After submission, the tree appears on the map for all users.

[Screenshot: Capture Form]

4.3 Explore

A social feed of recently spotted trees, sorted by time. Each card shows the tree photo, species, health, height, confidence bar, mapper’s name, and linked song with a play button. A “**View on Map**” button zooms directly to any tree’s location on the map.



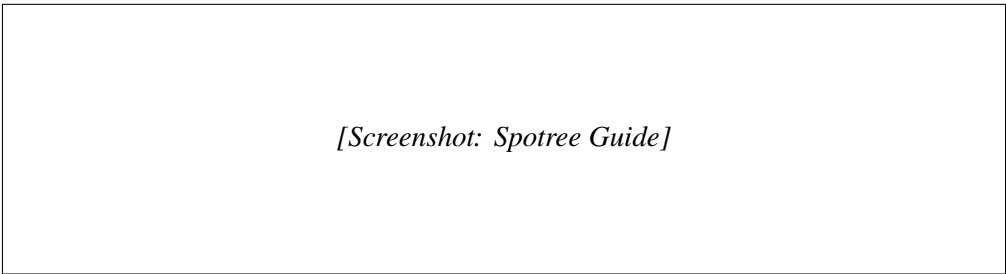
[Screenshot: Explore Feed]

4.4 Spotree Guide

An educational reference for Munich's native tree species:

- **12 species** with common and scientific names
- **Swipeable image galleries** with multiple photos per species
- **Fullscreen image viewer** for detailed examination
- Plant description chips: height, width, growth rate, flower color, blooming season, leaf retention
- Growth requirements: sun exposure, drainage needs
- Natural habitat description and common uses
- **External link to Easyscape** for deeper botanical learning

Data sourced from Easyscape, a botanical reference platform with verified species information.

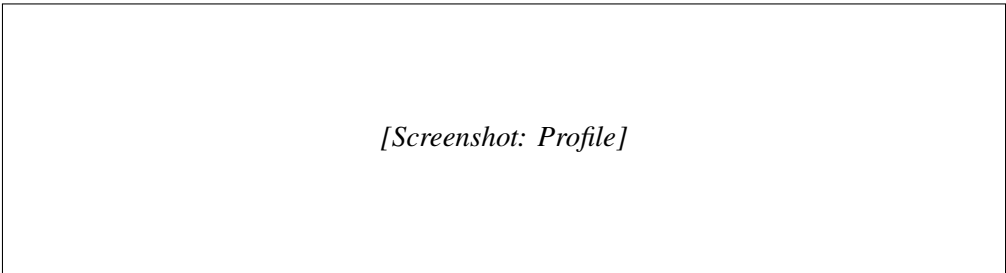


[Screenshot: Spotree Guide]

4.5 Profile

Personal dashboard showing:

- **Trees Spotted** — total count
- **Songs Shared** — trees with linked songs
- **Species Found** — unique species identified
- List of all personally mapped trees with dates and health status
- School affiliation (set once during registration)



[Screenshot: Profile]

4.6 Admin Analytics Dashboard

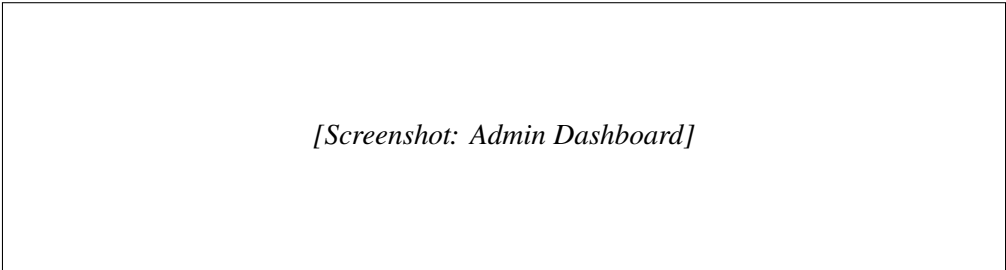
The admin dashboard is a web application designed for teachers, city administrators, and GeodataService staff to monitor the project at a glance. It connects to the same database as the mobile app and displays live data.

Layout: Stats panel on the left, full interactive map on the right — no scrolling needed. Grafana-inspired dark theme for comfortable monitoring.

Panels:

- **Stats Strip** — Total trees, registered users, unique species found, songs linked (with weekly deltas)
- **Species Breakdown** — Ranked list of all identified species with distribution bars
- **School Analytics** — Trees per school, showing which schools are most actively participating
- **Health Overview** — Distribution of tree health statuses across all observations
- **Activity Feed** — Real-time stream of the latest tree submissions with timestamps, species, and linked songs
- **Interactive Map** — All trees plotted on a dark-themed map with popups showing full details

School-level analytics are particularly valuable: they show which schools and neighborhoods are actively engaged, help identify coverage gaps, and enable targeted outreach to underrepresented areas.



[Screenshot: Admin Dashboard]

5. Data Collection & Value for GeodataService Munich

Every tree entry in Spotree captures a comprehensive dataset designed to be directly useful for Munich's GeodataService:

Field	Type	Purpose
GPS Location	PostGIS Point (4326)	Precise tree position from camera GPS or EXIF metadata
Photo	Image URL	Visual verification of tree existence and condition
Species	Text (58 natives)	Tree species identification
Species Confidence	Integer (0–100%)	Self-assessed certainty — signals data quality
Health Status	Enum	Current tree condition (Healthy/Fair/Poor/Dead)
Health Confidence	Integer (0–100%)	Certainty of health assessment
Estimated Height	Text (ranges)	Approximate tree height
Height Confidence	Integer (0–100%)	Certainty of height estimate
Trunk Diameter	Text (ranges)	Approximate diameter at breast height
Diameter Confidence	Integer (0–100%)	Certainty of diameter estimate
Notes	Free text	Additional observations (damage, surroundings, etc.)
Timestamp	ISO 8601	When the observation was recorded
User & School	Reference	Who collected the data and from which school

The **confidence sliders are a key design decision**. Rather than forcing students to guess, they can honestly mark uncertainty. This gives GeodataService a built-in quality metric: a tree with 90% species confidence from three students is far more reliable than a single observation at 30%. This enables quality-weighted data integration.

School-level tracking allows GeodataService to understand geographic coverage patterns — which neighborhoods are well-mapped based on which schools participate. This helps identify gaps and target outreach.

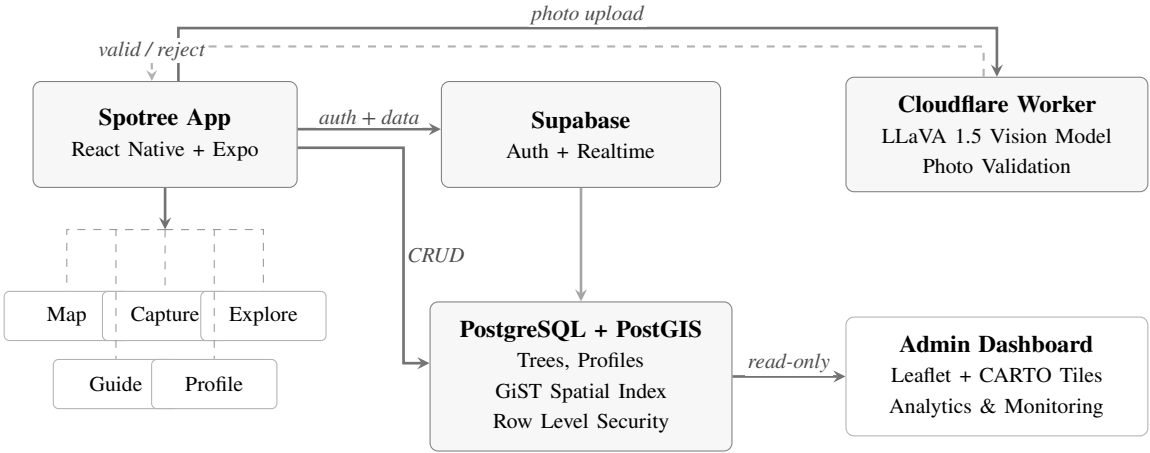
6. AI Integration

Crowdsourced data is only as valuable as its quality. To prevent misuse and ensure every database entry genuinely represents a tree, the prototype integrates AI-powered photo validation directly into the capture flow. When a student submits a photo, it passes through a lightweight on-device preprocessing step before being analyzed by a vision-language model on the server side:

1. Student takes or selects a photo
2. Photo is resized (512px) and base64-encoded on-device
3. Sent to a **Cloudflare Worker proxy** (server-side, no API keys exposed to client)
4. The Worker calls **LLaVA 1.5**, a vision-language model, which generates a natural language description of the image
5. Description is checked for tree-related keywords (tree, trunk, bark, branch, canopy, etc.)
6. Non-tree images are rejected with a clear message

This pipeline runs in under two seconds and requires no additional hardware on the student's device. During the co-creation phase, we plan to extend this with AI-powered species classification and continuous learning from student-submitted data (see Section ??).

7. Technical Architecture



7.1 Tech Stack

Component	Technology
Mobile App	React Native + Expo SDK 54, TypeScript
Authentication	Supabase Auth (email/password)
Database	PostgreSQL + PostGIS (via Supabase)
Spatial Data	geography(Point, 4326) with GiST spatial index
AI Validation	Cloudflare Workers AI — LLaVA 1.5 vision-language model
AI Proxy	Cloudflare Worker (keeps API keys server-side)
Image Processing	expo-image-manipulator (resize/compress before AI)
Maps (App)	react-native-maps (Apple Maps / Google Maps)
Maps (Admin)	Leaflet.js + CARTO dark tile layer
Admin Dashboard	Single-file HTML/JS — zero build step

8. OGC Compliance & Data Export

All tree data is stored in a **PostGIS** database using the standard `geography(Point, 4326)` type — WGS84 coordinate reference system, the global standard for GPS data and OGC-compliant spatial systems.

This means the data can be exported directly in standard GIS formats:

Format	Method	Use Case
GeoJSON	ST_AsGeoJSON() or API	Web mapping, API integration, lightweight exchange
Shapefile	pgsql2shp or QGIS	Traditional GIS software (ArcGIS, QGIS), municipal systems
GeoParquet	ogr2ogr with Parquet driver	Modern columnar format for large-scale spatial analytics
WFS/WMS	GeoServer or pg_featureserv	OGC web services for SDI infrastructure integration

The database schema uses a **spatial index** (GiST) on the tree locations for efficient spatial queries — bounding box lookups, nearest-neighbor searches, and spatial joins with existing Munich geodata layers.

Integration path: Munich’s GeodataService can connect directly to the PostGIS database via standard drivers, or consume exported files in their preferred format. The `trees_with_user` view provides a clean, denormalized dataset ready for analysis.

9. Co-Creation Phase Roadmap

The current prototype demonstrates the core concept and technical feasibility. The co-creation phase will expand Spotree into a production-ready platform.

9.1 Gamification & Reward System

Leaderboard: A public ranking of top contributors by trees spotted, species identified, and data quality (based on confidence scores and verification by other students). Visible in-app to drive friendly competition.

Area Ownership: Students can “own” a neighborhood zone (by postal code) by being the top contributor in that area. If another student maps more trees or provides higher-quality data in the same zone, ownership transfers. This creates ongoing engagement — students return to defend their territory and fill in gaps. It directly addresses the challenge goal of strengthening neighbourhood connection: students develop a sense of responsibility for “their” zone’s tree data.

Badges & Milestones: Unlock achievements for first tree, first species ID, mapping in multiple neighborhoods, seasonal mapping (capturing the same tree across seasons), etc.

9.2 AI-Powered Species Classification

The current prototype uses AI to validate that photos contain trees. The co-creation phase will add **automatic species identification**:

9.2.1 Approach

1. **Ground truth collection** — Use the prototype to collect GPS-tagged, species-labeled photos from students. Each photo has a species label and confidence score, giving us a quality-weighted training dataset.
2. **Seasonal variation** — Munich trees look significantly different across seasons (leaf-on vs. leaf-off, flowering periods, autumn colors). We will systematically collect images across all four seasons. A model trained only on summer photos would fail in winter.
3. **Munich-specific training** — Urban trees in Munich may look different from the same species in a forest or another climate zone. Local training data ensures the model reflects actual conditions.
4. **Model options:**
 - Fine-tune an open-source vision model on Munich-specific data
 - Evaluate existing providers (Pl@ntNet API, iNaturalist model) for baseline accuracy
 - Focus custom training on species where existing models struggle
5. **In-app flow:** After the student takes a photo, the AI suggests a species with a confidence score. The student can accept, correct, or mark as “Not Sure.” Corrections feed back into the training loop.

9.2.2 Continuous Learning from Student Data

Every tree submitted through Spotree becomes a potential training sample. The species label paired with the student’s confidence score acts as a **soft label** — a submission with 95% confidence is a stronger training signal than one at 40%. As the database grows, we can:

- Filter high-confidence submissions (e.g., >80%) as reliable ground truth for model retraining
- Use cross-verified entries (same tree, same species from multiple students) as gold-standard labels

- Retrain or fine-tune the model periodically with the latest data, so it improves as more students contribute
- Track model accuracy over time by comparing AI predictions against student consensus

This creates a **virtuous cycle**: students collect data, data trains the model, the model helps future students identify species more accurately, and those students submit even better-labeled data. The more Spotree is used, the smarter it gets.

9.3 Location Accuracy & Camera-to-Tree Distance

A photo's GPS tag records *where the camera was*, not where the tree is. If a student photographs a tree from 20 meters away, the mapped coordinate could be significantly offset from the actual trunk location. For GeodataService Munich, positional accuracy is critical — duplicate detection, cross-referencing with existing tree cadastres, and spatial joins all depend on coordinates being close to the true tree position.

The co-creation phase will investigate and implement a methodology for ensuring location accuracy:

9.3.1 Distance Estimation

- **Monocular depth estimation** — Use a lightweight depth model (e.g., MiDaS or Depth Anything) to estimate the camera-to-tree distance from a single photo. The tree trunk can be identified as the primary foreground object, and its relative depth converted to an approximate metric distance using known camera parameters (focal length from EXIF).
- **Object size heuristics** — If the student has already estimated trunk diameter or tree height, these physical dimensions combined with their pixel size in the image provide a second distance estimate via basic photogrammetry.
- **Threshold enforcement** — Define a maximum acceptable camera-to-tree distance (e.g., 5–10 meters). If the estimated distance exceeds this threshold, prompt the student: *“You seem far from the tree — please move closer for a more accurate location.”* This nudge improves data quality at the point of capture.

9.3.2 Coordinate Correction

- **Bearing-based offset** — Using the device compass heading at capture time and the estimated distance, compute a corrected coordinate that shifts the GPS point from the camera position toward the tree along the viewing direction.
- **Multi-observation averaging** — When multiple students photograph the same tree from different positions, triangulating their corrected coordinates produces a more accurate consensus location.
- **Accuracy metadata** — Store the estimated camera-to-tree distance and correction method as metadata alongside each tree entry, giving GeodataService a per-record quality indicator for positional accuracy.

This investigation is directly relevant to the team's expertise: Jayendra's work on geospatial data pipelines and Hari's research on image-based geolocation (ICLR 2026) provide the methodological foundation for building robust distance estimation and coordinate correction into Spotree.

9.4 Educational Earth Observation Module

A new tab in the Spotree Guide showing **how tree species appear from different sensors and perspectives**:

- **High-resolution satellite imagery** — What does a cluster of this species look like from orbit? (Using freely available Sentinel-2 or commercial VHR imagery)
- **Hyperspectral views** — How does the spectral signature of this species differ from others? This

introduces students to the concept that sensors can “see” beyond visible light

- **LiDAR point clouds** — 3D structure of tree canopies, connecting to the height and diameter measurements students make on the ground
- **Seasonal comparisons** — Same location across seasons from satellite, showing phenological changes

This bridges the gap between ground-level observation and remote sensing, introducing students to how cities actually monitor urban forests at scale. Space and satellite technology is inherently exciting for students and adds a unique educational dimension beyond traditional biology lessons.

9.5 Platform Scalability

The Spotree architecture is **not limited to trees**. The same capture-validate-map-explore pattern can be extended to other urban assets:

- **Public benches** — location, condition, accessibility
- **Table tennis tables** — public sports infrastructure mapping
- **Playgrounds** — equipment inventory and condition

Each asset type gets its own capture form, guide section, and map layer. The database schema, admin dashboard, and gamification system remain the same. This makes Spotree a general-purpose **crowdsourced urban data collection platform** that starts with trees and grows with the city’s needs.

9.6 Production Infrastructure

The current prototype runs on Supabase’s free tier. For production deployment:

Component	Technology	Hosting
Mobile App	React Native (iOS + Android)	App Store / Play Store
App Backend API	Dedicated service	Hetzner Cloud (German-based)
Database	PostgreSQL + PostGIS	Hetzner Cloud
DB Admin	pgAdmin	Hetzner Cloud
Analytics	Grafana (connected to PostGIS)	Hetzner Cloud
AI Service	Dedicated inference service	Hetzner Cloud / Cloudflare

Why Hetzner: German-based hosting ensures GDPR compliance with data residency in Germany. No dependency on external SaaS providers — full control over the data pipeline. Microservice architecture allows independent scaling of each component.

Why self-hosted Grafana: Professional-grade monitoring with alerting, time-series analysis, custom queries, and multi-user access control. City staff can build their own dashboards without developer involvement.

During the co-creation phase, we will actively visit Munich schools to present and demonstrate Spotree, ensuring students and teachers across the city know about the app and how to use it for real data collection.